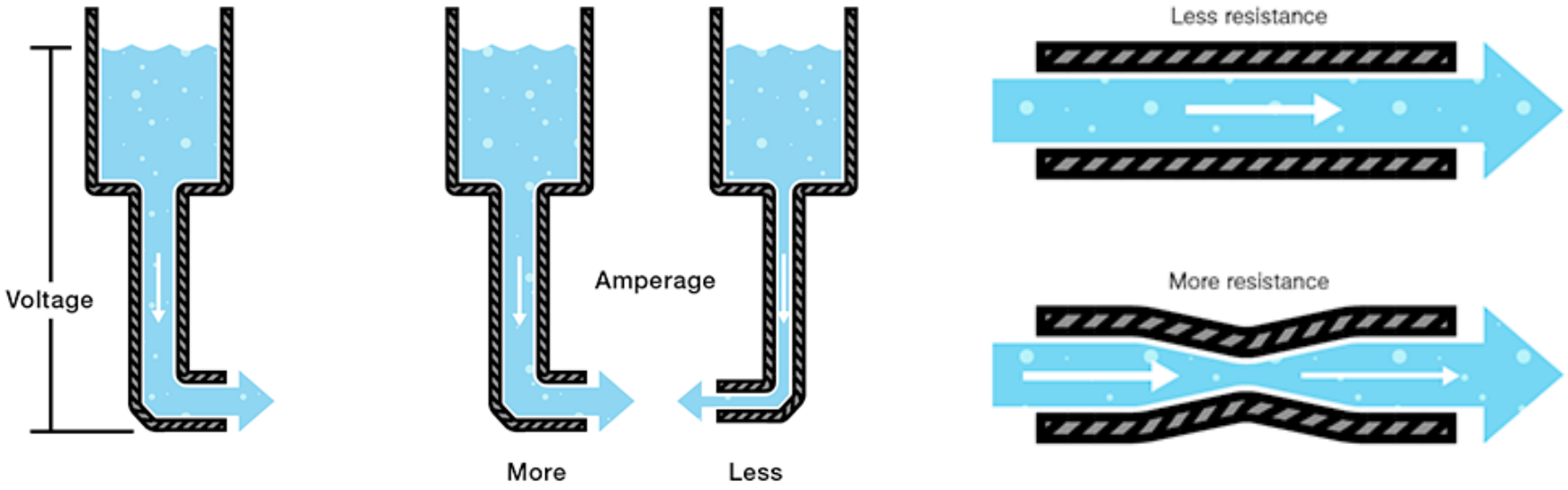
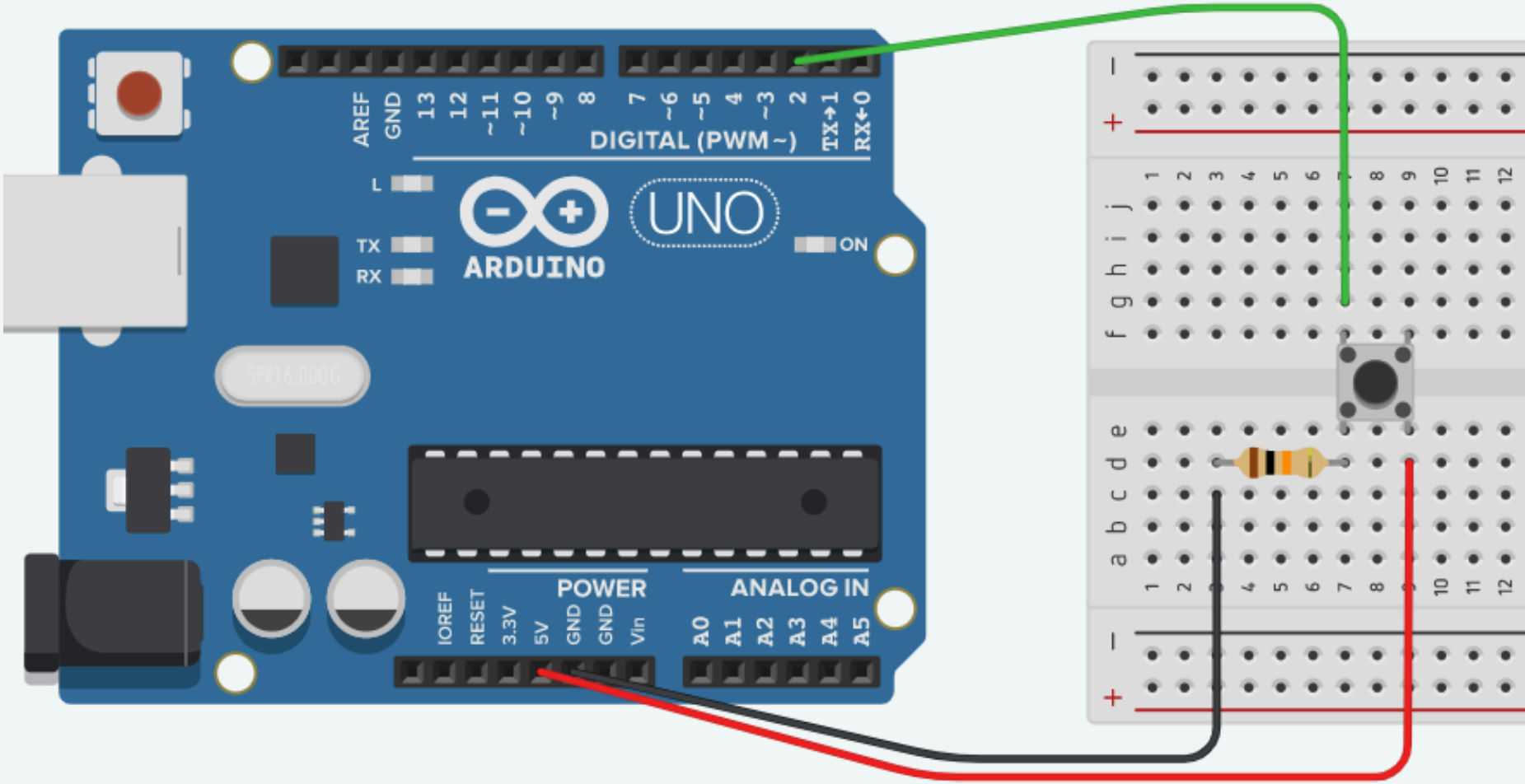
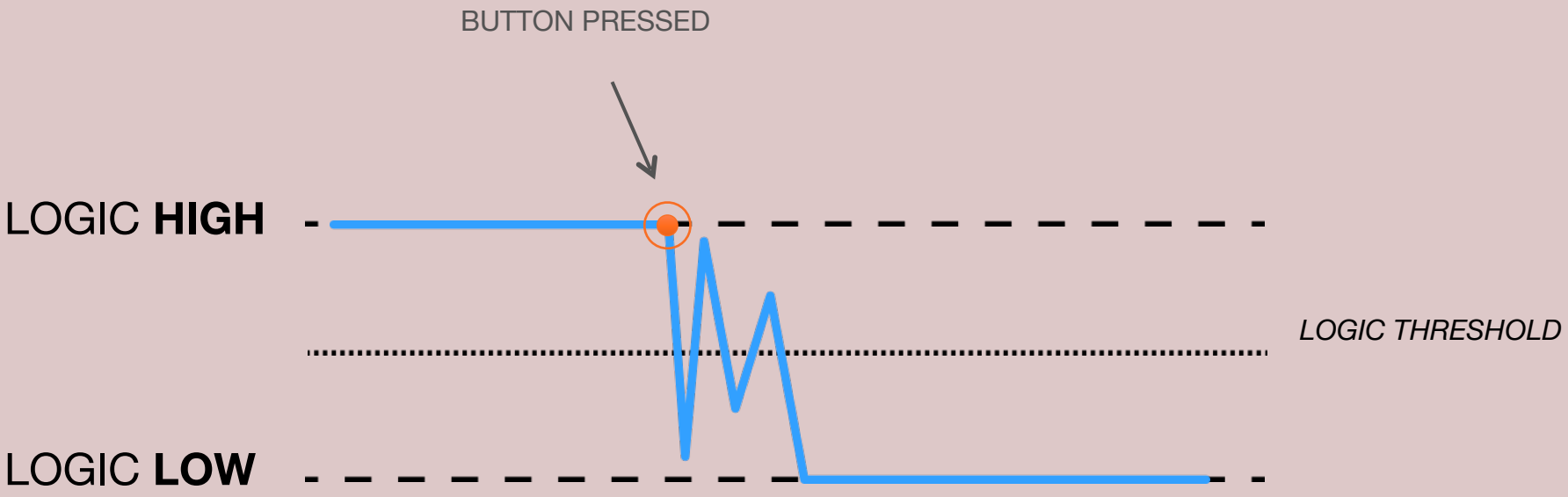
TECNOLOGIAS DE INTERFACE

ANO LETIVO 2022/2023


Voltage
Amperage
More
Less
Less resistance
More resistance

$$I = \frac{V}{R} \quad \text{or} \quad V = IR \quad \text{or} \quad R = \frac{V}{I}.$$


AREF
GND
13
12
~11
~10
~9
8
7
~6
~5
4
~3
2
TX→1
RX←0
DIGITAL (PWM~)
L
TX
RX
ARDUINO
UNO
ON
IOREF
RESET
3.3V
5V
GND
GND
Vin
POWER
ANALOG IN
A0
A1
A2
A3
A4
A5


BUTTON PRESSED
LOGIC HIGH
LOGIC THRESHOLD
LOGIC LOW

How to avoid Bouncing?

Connecting a pushbutton to the Arduino with **debouncing**

1. Read the state of the button
2. Check if you pressed the button. if the button state changes, reset debounce timer
3. Check if the timer is superior to our debouncing delay. if it is we assume the current state reading
4. Save the reading for the next loop

Pushbutton_debouncing | Arduino IDE 2.0.3

Arduino Uno

Pushbutton_debouncing.ino

```
const int BT_PIN = 2;    // pushbutton pin
const int LED_PIN = 13;  // led pin

int btState = 0;             // variable for reading the pushbutton status
int ledState = HIGH;         // current state of the led
int previousLedState = LOW;  // previous state of the led

// time variables
long lastDebounceTime = 0;   // the last time the output pin was toggled
long debounceTimeDelay = 0;  // the bounce time;


void setup() {
  // init pins
  pinMode(LED_PIN, OUTPUT);
  pinMode(BT_PIN, INPUT);
}

void loop() {
  // read the state of pushbutton pin
  int reading = digitalRead(BT_PIN);

  // check to see if you just pressed the button
  // (i.e. the input went from LOW to HIGH), and you are waited
  // long enough since the last press to ignore any noise:.

  // If the switch changed, due to noise or pressing:
  if (reading != lastDebounceTime) {
    // reset the debouncing timer
    lastDebounceTime = millis();
  }

  if ((millis() - lastDebounceTime) > debounceTimeDelay) {
    // whatever the reading is at, it is been there for longer
```

Output

```
Sketch uses 1032 bytes (3%) of program storage space. Maximum is 32256 bytes.
Global variables use 15 bytes (0%) of dynamic memory, leaving 2033 bytes for local variables. Maximum is 2048 bytes.
```

Ln 57, Col 1 UTF-8 Arduino Uno on /dev/cu.usbmodem2101 1

Code: https://github.com/sergiomrebelo/fctuc-mdm-ti/tree/main/PL2-Pushbutton

Aula prática 3

Arduino & Friends:
Going Further **(part 2)**

$$I = \frac{V}{R} \quad \text{or} \quad V = IR \quad \text{or} \quad R = \frac{V}{I}.$$

An LED needs about 2-3V and 20-25 mA, so since we are putting 5V across it from the Arduino, we should place a resistor in series with the LED. According to Ohm's Law, we need a 220 Ohm resistor (red red brown).

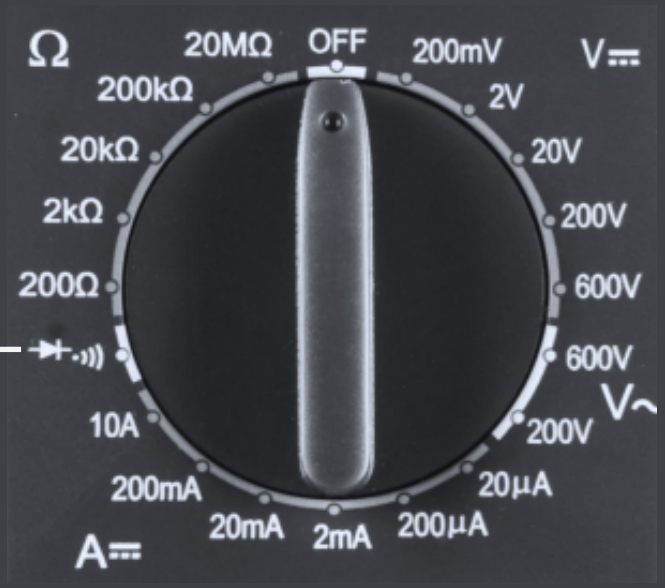
Continuity

Is it a Circuit?

An Electrical Circuit must have a continuous, closed loop from Power (VCC) to Ground (GND).

Continuity is important to make portions of circuits are connect. Continuity is the basic and one of the most important settings on your multi-meter.

CONTINUITY
SETTING



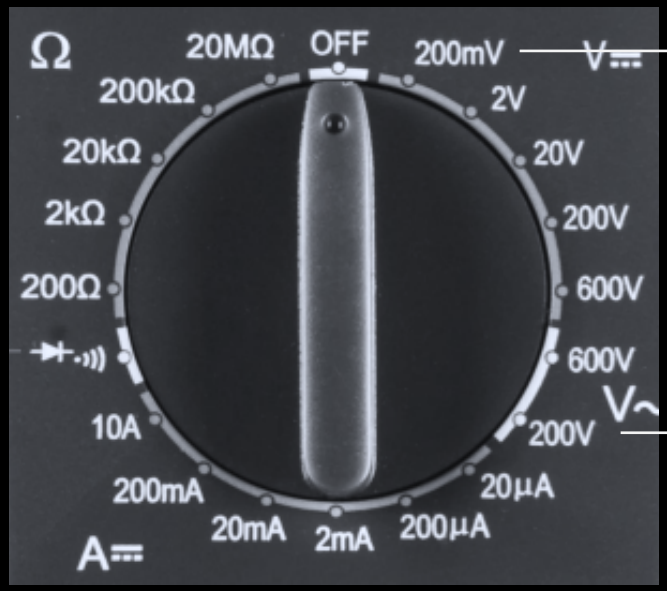
Measuring Electricity

Measuring Electricity — **Voltage**

Measuring Electricity — **Voltage**

Voltage is a measure of potential electrical energy.

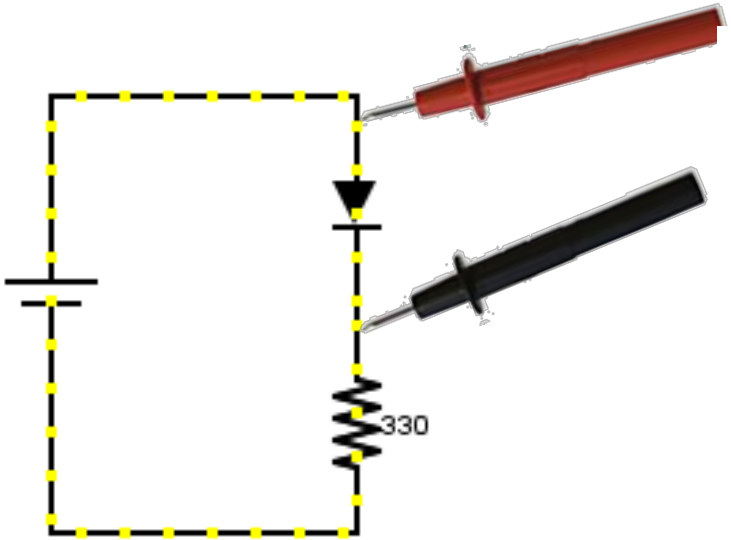
A voltage is also called a potential difference — it is measured between two points in a circuit — across a device.

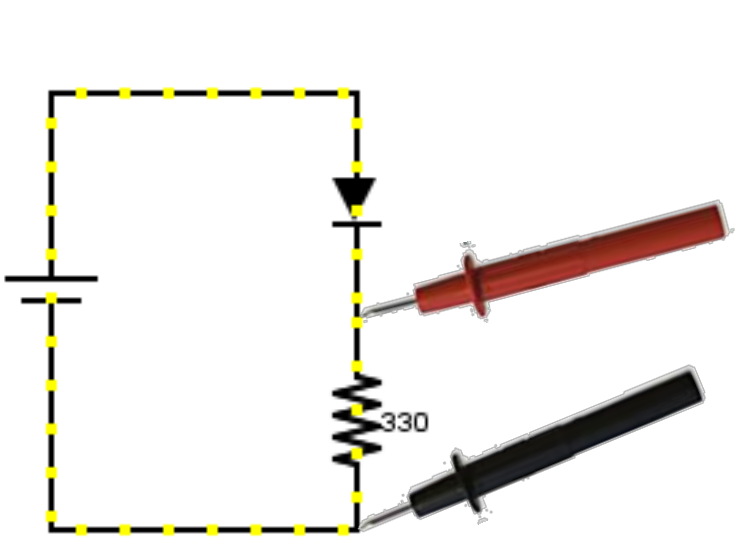


LOWEST VALUE

VOLTAGE SETTINGS

HIGHEST VALUES


330

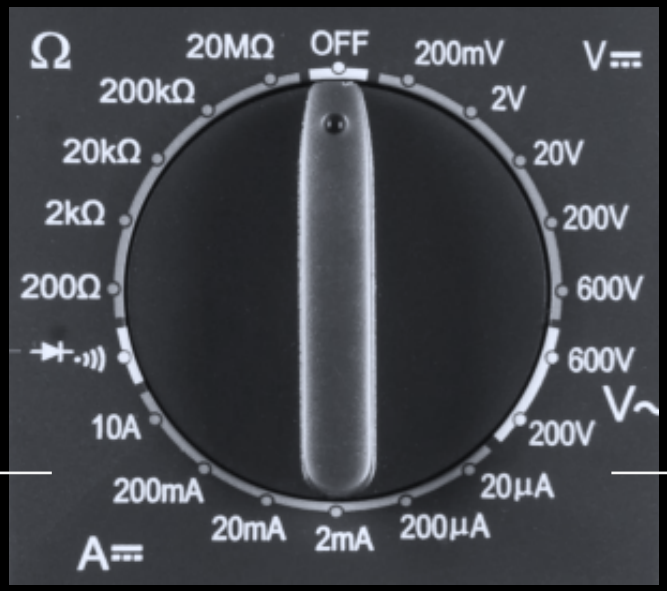

330

Measuring Electricity — **Current**

Measuring Electricity — **Current**

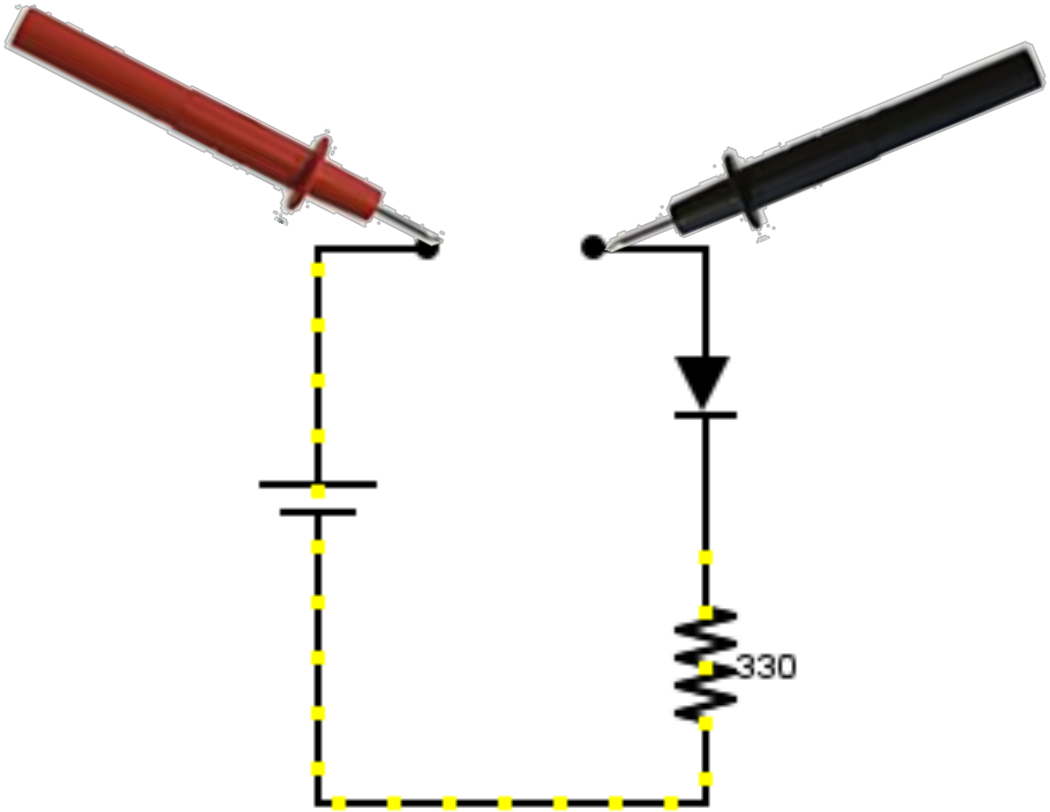
Current is the measure of the rate of charge flow.
For Electrical Engineers — we consider this to be the movement of electrons.

In order to measure this — you must break the circuit or insert the meter in-line (series).



HIGHEST VALUES ← → LOWEST VALUE

CURRENT SETTINGS


330

Measuring Electricity — **Resistance**

Measuring Electricity — **Resistance**

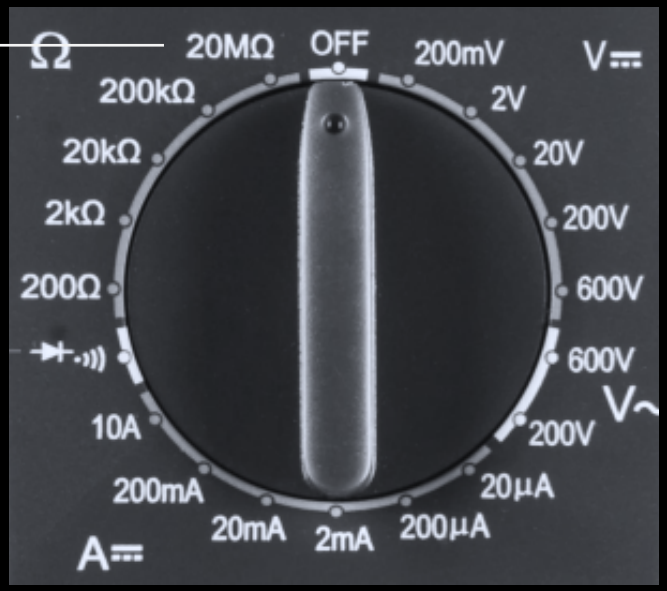
Resistance is the measure of how much opposition to current flow is in a circuit. Components should be removed entirely from the circuit to measure resistance!

*Note the settings on the multi-meter. Make sure that you are set for the appropriate range.

HIGHEST VALUE

RESISTANCE SETTINGS

LOWEST VALUE



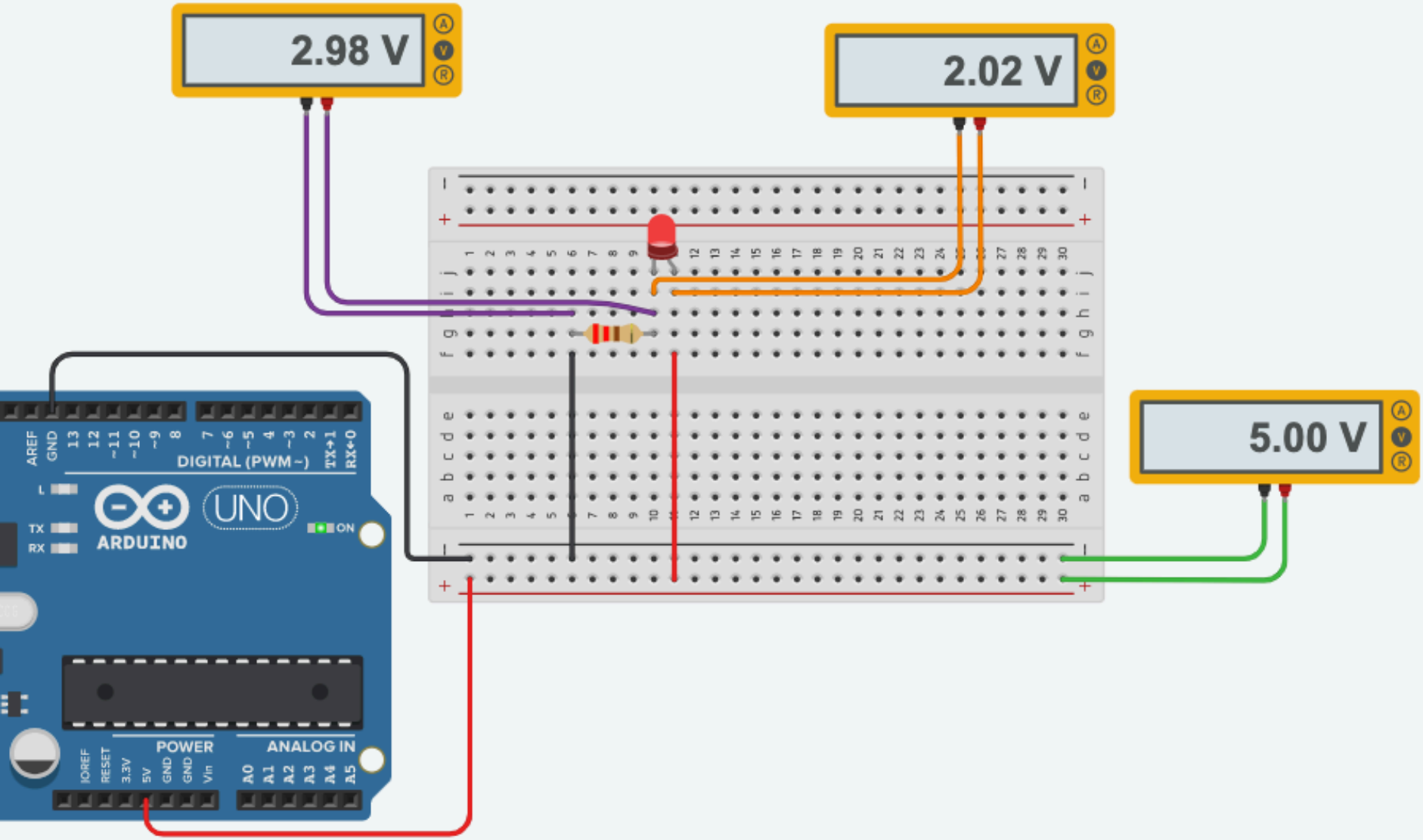
Exercise #1

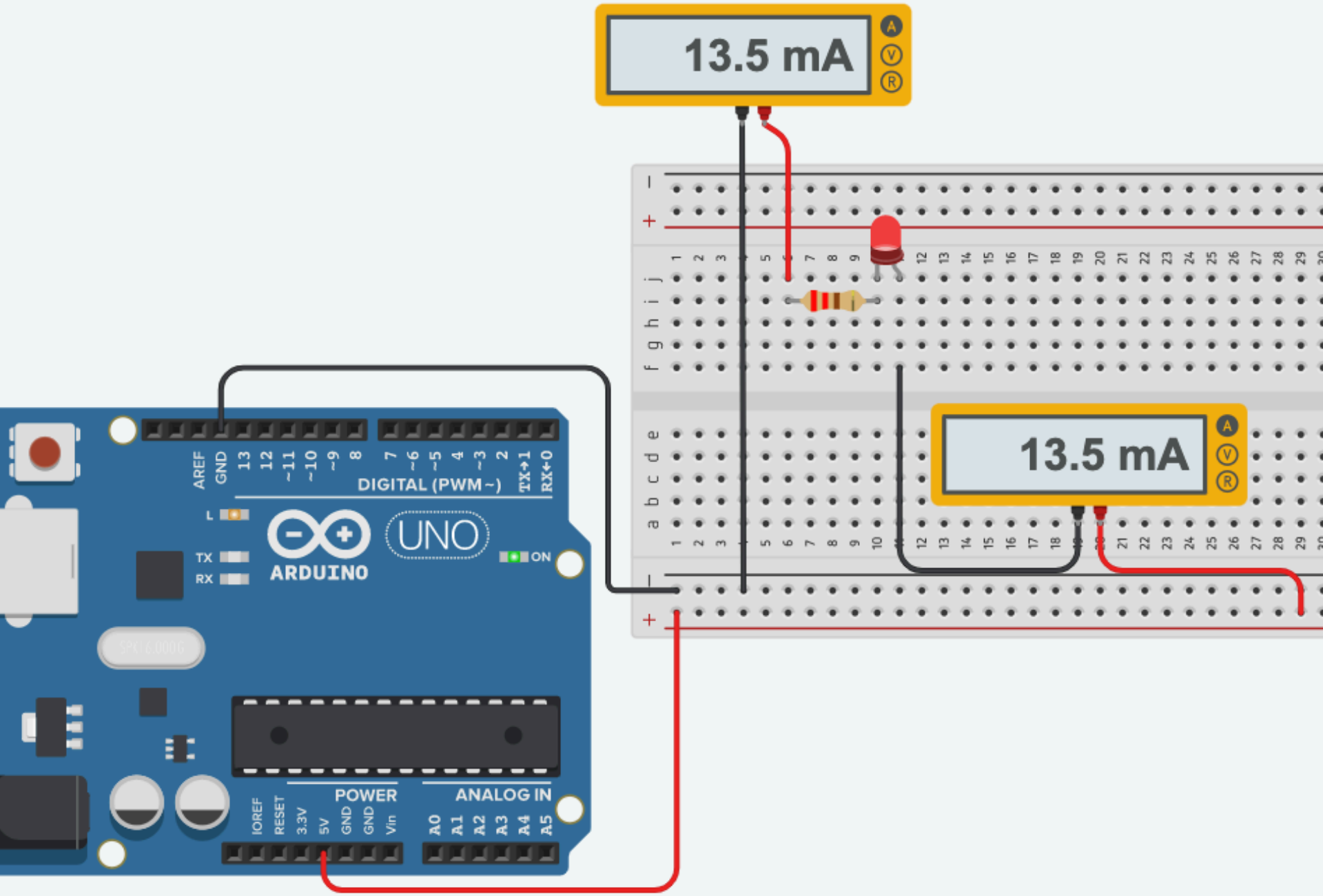
1. Pick a 220 Ohm resistor. Measure it with the multi-meter.

2. Connect a LED between 5V and GND, using a 220 Ohm resistor.

3. Measure the voltage across the LED and the resistor.
   Now measure the voltage between the +5V and GND pins.

4. Measure the current on the circuit, on two different places.

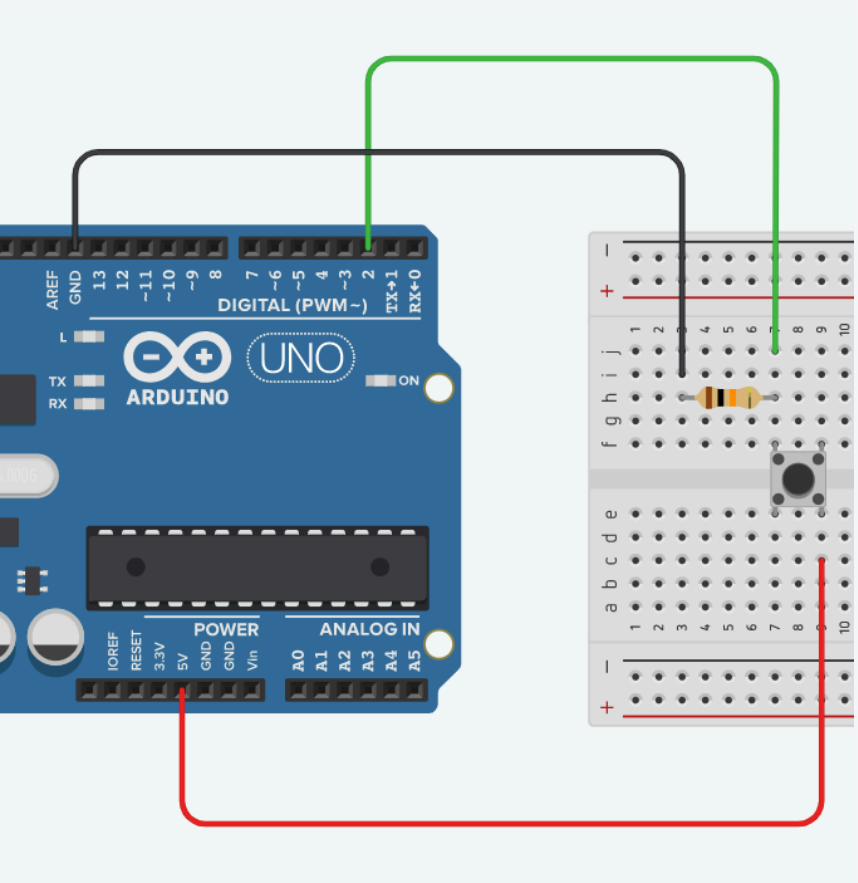

2.98 V
2.02 V
5.00 V
AREF
GND
13
12
~11
~10
~9
8
7
~6
~5
4
~3
2
TX→1
RX←0
DIGITAL (PWM~)
L
TX
RX
UNO
ARDUINO
ON
IOREF
RESET
3.3V
5V
GND
GND
Vin
POWER
ANALOG IN
A0
A1
A2
A3
A4
A5


13.5 mA
13.5 mA
AREF
GND
13
12
~11
~10
~9
8
7
~6
~5
4
~3
2
TX→1
RX←0
DIGITAL (PWM~)
L
TX
RX
ARDUINO
UNO
ON
IOREF
RESET
3.3V
5V
GND
GND
Vin
POWER
ANALOG IN
A0
A1
A2
A3
A4
A5

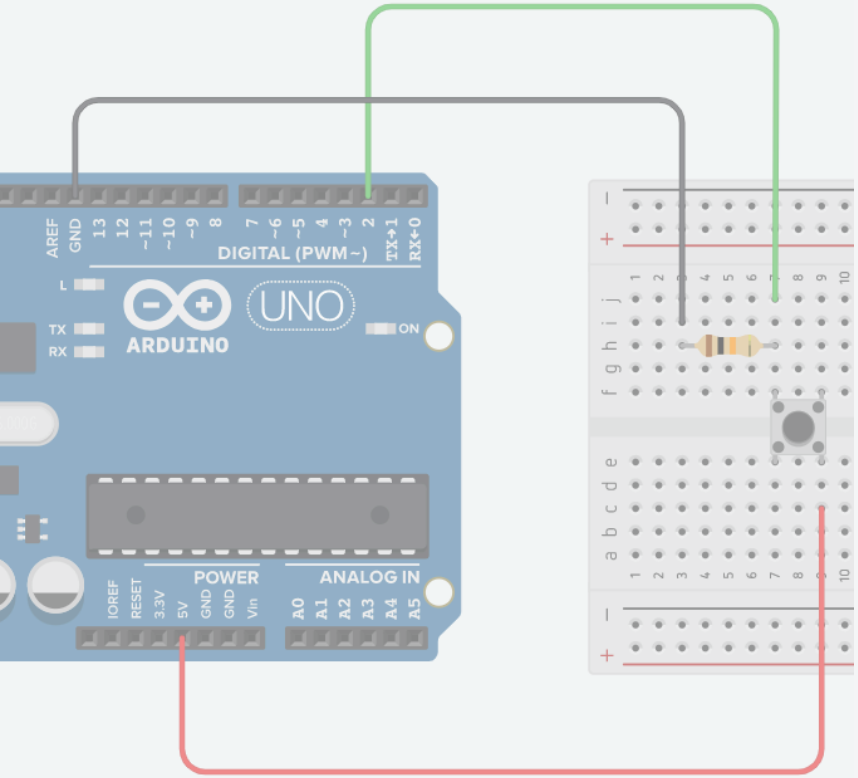
Exercise #2

Write down all the readings and check if Ohm’s Law holds true!

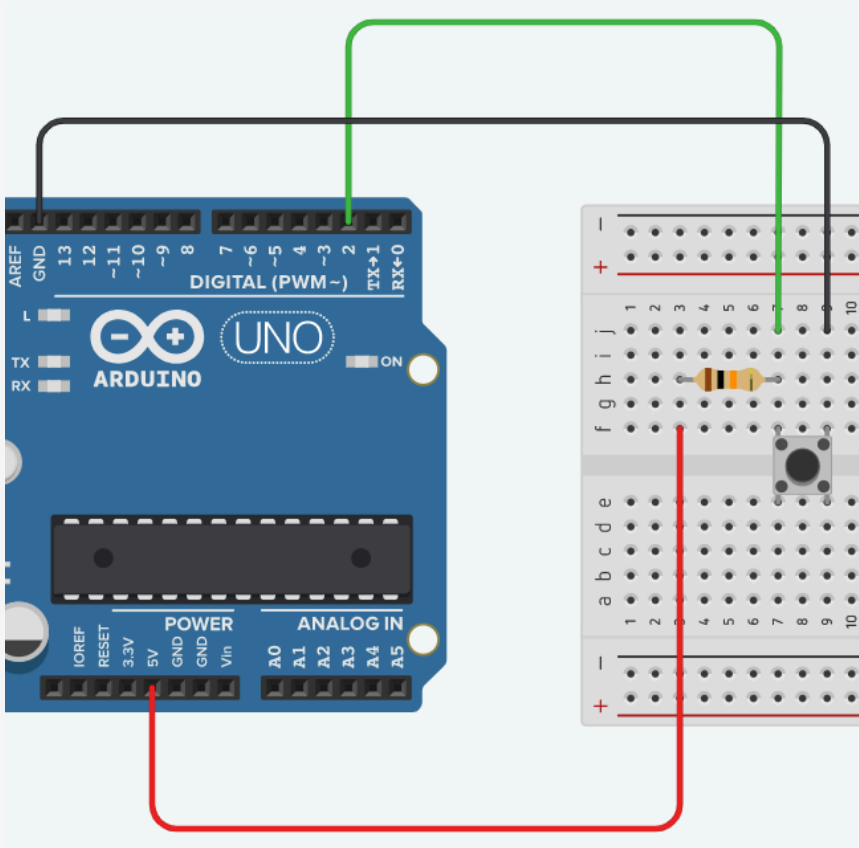
Pull-ups vs **pull-downs**



Pull-down Pushbutton

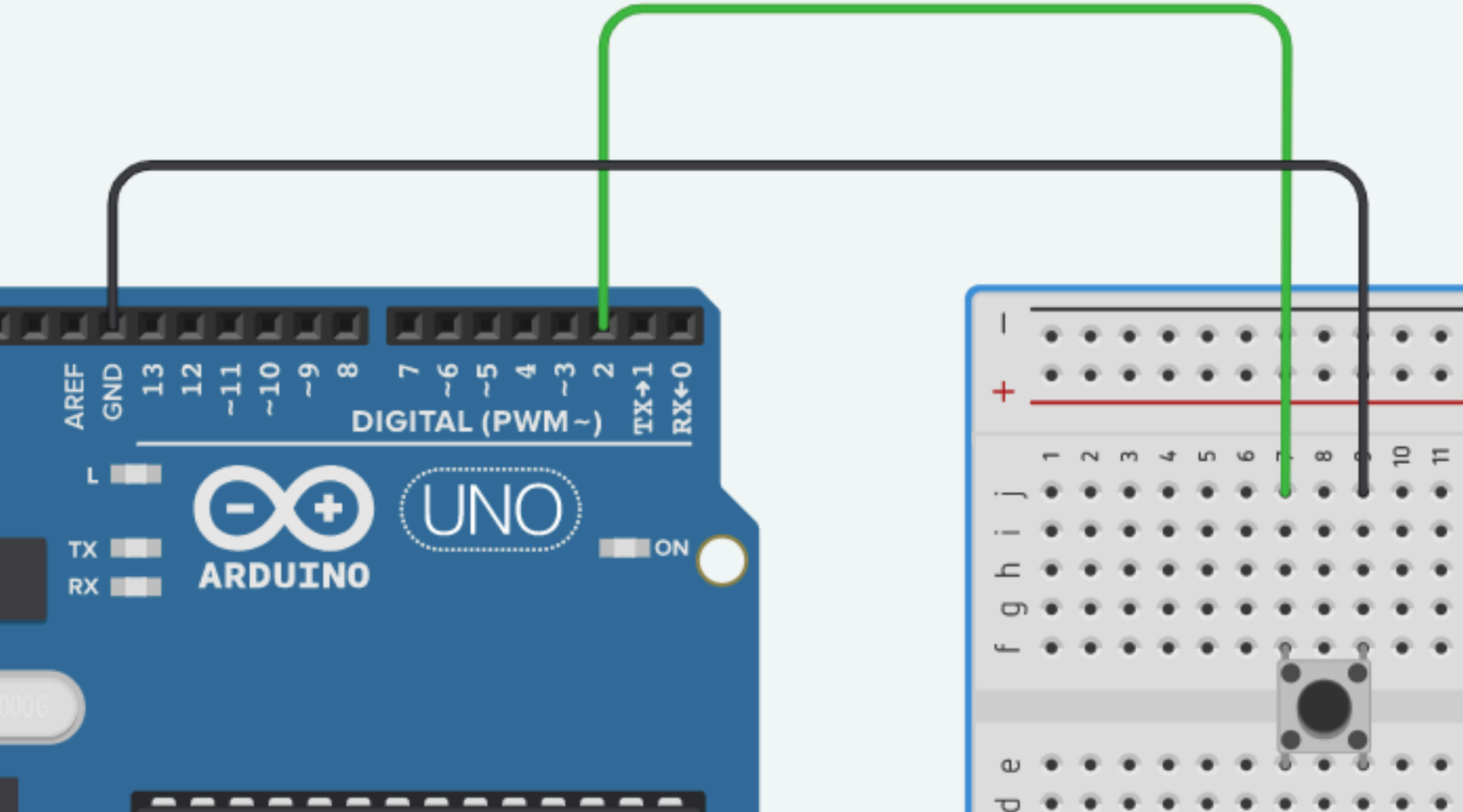


Pull-down Pushbutton



Pull-up Pushbutton

The Arduino has internal pull-ups! By declaring a pin as **INPUT_PULLUP** (instead of **INPUT**). You do not need external resistors anymore!


AREF
GND
13
12
~11
~10
~9
8
7
~6
~5
4
~3
2
TX→1
RX←0
DIGITAL (PWM~)
L
TX
RX
ARDUINO
UNO
ON

Week3_Pushbutton_pullup | Arduino IDE 2.0.3

Arduino Uno

Week3_Pushbutton_pullup.ino

```
// time variables
long lastDebounceTime = 0;   // the last time the output pin was toggled
long debounceTimeDelay = 0;  // the bounce time;


void setup() {
  // init pins
  pinMode(LED_BUILTIN, OUTPUT);
  pinMode(BT_PIN, INPUT_PULLUP);
}

void loop() {
  // read the state of pushbutton pin
  int reading = digitalRead(BT_PIN);

  // check to see if you just pressed the button
  // (i.e. the input went from LOW to HIGH), and you are waited
  // long enough since the last press to ignore any noise:.

  // If the switch changed, due to noise or pressing:
  if (reading != lastDebounceTime) {
    // reset the debouncing timer
    lastDebounceTime = millis();
  }

```

Output

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Ln 27, Col 31 UTF-8 Arduino Uno on /dev/cu.usbmodem101 2

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